

University of the Philippines-Diliman
Department of Computer Science

CS 210
First Long Exam

11 April 2015
A.R.L. Valdez

DIRECTION: Answer all problems clearly and completely specifying all justifications for arguments which are not obvious (to a 5 year old kid). Good luck!

1. Solve the following recurrences:

a) $n(n-1)a_n = (n-1)a_{n-1} + a_{n-2}$ for $n \geq 2$ with $a_0 = a_1 = 1$. [4pts]

b) $a_n = \sqrt{1 + a_{n-1}^2}$ for $n \geq 1$ with $a_0 = 0$. [4pts]

2. Consider the initial value problem:

(1) $zf'(z) - zf(z) = (e^z - 1), \quad f(0) = 0.$

a) Show that $f(z) = e^z \int_0^z \frac{1 - e^{-t}}{t} dt$ solves (1). [3pts]

b) Show that the EGF $\mathcal{H}(z)$ of the Harmonic numbers H_n solves (1). [3pts]

c) Find a simple expression for $N![z^N] e^z \int_0^z \frac{1 - e^{-t}}{t} dt$. [1pt]

3. Given the recurrence relation

$$a_n = 6a_{n-1} - 4a_{n-2} - 6a_{n-3} + 5a_{n-4}, \quad \text{for } n \geq 4.$$

a) What is $g(z)$ as defined in Theorem 3.3 of the notes (page 104) expressed as $g(z) = 1 - x_1z - \dots x_tz^t$? [1pt]

b) Express $g(z)$ as a product of linear factors of the form $(1 - \beta z)$ for $\beta \in \mathbb{R}$ with $(1 - 5z)$ being one of them. [2pts]

c) With $a_0 = 2, a_1 = 4, a_2 = 26, a_3 = 124$, find $f(z)$ as defined in Theorem 3.3. [2pts]

d) Express $f(z)$ as a product of linear factors of the form $(1 - \beta z)$ for $\beta \in \mathbb{R}$ with $(1 - 2z)$ being one of them. [2pts]

e) What is $f(z)/g(z)$ in simplest form? Express your answer in partial fractions. [2pts]

f) Solve for a_n using the given in (3.c). [1pt]

g) With $a_0 = a_1 = 1$ and $a_2 = a_3 = 2$, find $\hat{f}(z)$ as defined in Theorem 3.3. [2pts]

h) What is $\hat{f}(z)/g(z)$ in simplest form? [2pts]

i) Find an approximate value for $[z^n] \frac{\hat{f}(z)}{g(z)}$ using Theorem 4.1 (page 157). [2pts]

4. Asymptotically expand the following functions:

a) $e^{\frac{N}{N-1}}$ as $N \rightarrow \infty$ to within $O(N^{-4})$. [4pts]

b) $\frac{x}{\ln(1+x)}$ as $x \rightarrow 0$ to within $O(x^4)$. [4pts]

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5. Consider the following recurrence:

$$\begin{cases} a_{2n+1} &= 2a_n + \gamma \\ a_{2n} &= 2a_n + \beta \\ a_1 &= \alpha. \end{cases}$$

- a) Make a table for the recurrence values of a_n for $n = 1, \dots, 10$. [3pts]
b) Solving the recurrence above using the repertoire method, let us assume that a_n is of the form:

$$(2) \qquad a_n = \alpha A(n) + \beta B(n) + \gamma C(n).$$

Looking at the table in (5.a), what is a possible candidate for $A(n)$? Express your answer as a function of $m := \lfloor \log_2 n \rfloor$.

Note: $n = 2^m + k$ for some $0 \leq k < 2^m$. [1pt]

- c) For $a_n = 1$, determine α , β and γ and rewrite (2)? [1pt]
d) For $a_n = n$, determine α , β and γ and rewrite (2)? [1pt]
e) For $a_n = 2k + 1$, with k as defined in (5.b), determine α , β and γ and rewrite (2)? [2pts]
f) Using the answers obtained in (5.c)-(5.e), write the precise solution of a_n by solving the values of $A(n)$, $B(n)$ and $C(n)$. [3pts]

End of Examination: 50 points